1. **Title of the Study:** Design and Evaluation of a Human-Centred, AI-Assisted Platform for Transforming Real-World Problems into Software Requirements and Supporting Their Implementation

2. **Background of the study**

   Many useful software projects start with people who understand a problem well but cannot write it as software requirements. Incomplete or hidden requirements and weak communication between customers and developers remain among the most critical causes of failed projects (Méndez Fernández et al., 2017). Large language models can now draft requirements and proposals of reasonable quality (Ronanki et al., 2023), but they also produce confident statements that their input does not support (Ji et al., 2023).

   The Automation Platform, built with Django, GraphQL, React and PostgreSQL, already supports much of the path from a problem to delivered software. Owners describe their problem in plain language through a guided intake form. Review teams then assess it against five criteria, and a proposal is written and released to the owner. After the owner's go-ahead, the work is tracked through requirements, assignment, testing, deployment and impact. A first AI feature, which drafts single proposal sections for reviewers, has been built but not yet evaluated.

3. **Statement of the problem**

   Although the platform captures the owner's problem in detail, reviewers still turn that narrative into proposals and requirements by hand. Missing information is often discovered late, as change requests or placeholders. Once AI-drafted text is accepted it looks exactly like human-written text, so readers cannot tell what the owner said, what reviewers concluded and what the AI suggested. Estimates are not separated from verified figures, and requirements are not linked to the tasks and tests that implement them. Without evidence on how to add AI to this workflow safely, more automation could make documents look complete while making them harder to trust.

4. **Aim of the Study**

   This study aims to design, implement and evaluate an AI-assisted extension of the Automation Platform. The extension will help transform owners' problem descriptions into professional documents and traceable software requirements while preserving the owner's intent and keeping owners and reviewers in control of decisions. The study follows the workflow from idea submission and review through proposal writing and requirements, up to the development tasks and tests that implement them.

5. **Research Objectives**

   - **General objective**

     To develop and evaluate a human-centred, AI-assisted workflow that turns real-world problems into reliable software requirements and supports their

implementation.

- **Specific objectives**
  - To assess the existing platform and identify the information gaps in its intake, review and proposal workflow.
  - To design AI support that identifies missing information, asks the owner targeted clarification questions and drafts documents in which every statement shows its source.
  - To implement the extension, including links from approved requirements to development tasks and tests.
  - To evaluate the quality, intent preservation, correction effort and usability of AI-assisted documents with owners and reviewers.

6. Main Research Question

How can AI assistance be designed into a software automation platform so that non-technical descriptions of real-world problems become professional documents and traceable software requirements, without losing the owner's intent or human control over decisions?

7. Significance of the study

- Academic significance

  The study will add evaluated evidence on AI-assisted requirements engineering inside a real, multi-role workflow. Current reviews describe this evidence as scarce: research on language technology for requirements has produced many tools with little practical evaluation (Zhao et al., 2021), and AI research in software engineering has focused mainly on code (Hou et al., 2024).

- Practical significance

  The study could reduce the effort reviewers spend writing proposals and requirements. It could also help non-technical owners take part in the requirements process on equal terms, and make generated documents easier to check before money and developer time are committed.

- Potential beneficiaries

  Idea owners and their organisations, platform reviewers and delivery teams, developers, and researchers in software engineering and human-centred AI.

8. Scope of the Study

The study covers the path from an owner's submitted problem to approved, traceable requirements and their link to development tasks and tests. It focuses on gap identification, clarification questions, AI-assisted drafting of a concept note, the proposal and a requirements set, and a label on every statement showing its source. Autonomous code generation and deployment by AI, replacing the existing intake

form, and financial advice are outside the scope. A link from projects to code repositories is optional and will be attempted only if time allows.

9. Methodology

Research Design

The study will use design science research, which builds and evaluates an artefact that addresses a practical problem (Hevner et al., 2004). It will follow the six activities of the design science research methodology (Peffers et al., 2007), and the evaluation will be planned with the FEDS framework (Venable et al., 2016).

Data Collection

Data will come from four sources: an audit of the platform's code and documentation; a set of representative project scenarios with known missing facts; logs of the AI suggestions that users accept, edit or reject; and questionnaires and short interviews with idea owners and reviewers. The number of participants will depend on access and ethics approval.

Data Analysis

Documents written with and without AI assistance will be rated blind for completeness, clarity, consistency, correctness and preservation of the owner's intent. Usability will be measured with the System Usability Scale (Brooke, 1996). Quantitative data will be summarised with descriptive statistics and paired comparisons, and interview data will be analysed thematically.

Ethical Considerations

Ethics approval will be obtained before any participant data are collected. Participation will be voluntary and confidential, and real ideas will be used only with their owners' consent. Owners will be told when their text is sent to an external AI service, and personal data will be kept out of AI prompts.

10. Expected Outcomes

The study is expected to provide: an assessment of what the platform already does and where its information gaps lie; a tested prototype that identifies missing information, asks clarification questions, drafts documents with visible sources and links requirements to tasks and tests; evaluation results on document quality, intent preservation, correction effort and usability; and practical design principles for human-centred AI in requirements work. No results are claimed before the research is carried out.

11. Work Plan / Timeline

Activity	Proposed period
1. Literature review and assessment of the existing platform	Month 1
2. Requirements refinement, workflow design and ethics application	Month 2
3. Prototype implementation	Months 2–4
4. Technical testing and evaluation preparation	Month 4
5. Owner and reviewer evaluation, and data analysis	Month 5
6. Final report, conclusions and presentation	Month 6

The final timeline will be adjusted to ethics approval, participant availability and supervisor guidance.

References

Brooke, J. (1996). SUS: A “quick and dirty” usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability evaluation in industry* (pp. 189–194). Taylor & Francis.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly, 28*(1), 75–105. https://doi.org/10.2307/25148625

Hou, X., Zhao, Y., Liu, Y., Yang, Z., Wang, K., Li, L., Luo, X., Lo, D., Grundy, J., & Wang, H. (2024). Large language models for software engineering: A systematic literature review. *ACM Transactions on Software Engineering and Methodology, 33*(8), 1–79. https://doi.org/10.1145/3695988

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), Article 248. https://doi.org/10.1145/3571730

Méndez Fernández, D., Wagner, S., Kalinowski, M., Felderer, M., Mafra, P., Vetrò, A., et al. (2017). Naming the pain in requirements engineering: Contemporary problems, causes, and effects in practice. *Empirical Software Engineering, 22*(5), 2298–2338. https://doi.org/10.1007/s10664-016-9451-7

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Ronanki, K., Berger, C., & Horkoff, J. (2023). Investigating ChatGPT’s potential to assist in requirements elicitation processes. In *2023 49th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)* (pp. 354–361). IEEE. https://doi.org/10.1109/SEAA60479.2023.00061

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A framework for evaluation in design science research. *European Journal of Information Systems, 25*(1), 77–89. https://doi.org/10.1057/ejis.2014.36

Zhao, L., Alhoshan, W., Ferrari, A., Letsholo, K. J., Ajagbe, M. A., Chioasca, E.-V., & Batista-Navarro, R. T. (2021). Natural language processing for requirements engineering: A systematic mapping study. *ACM Computing Surveys, 54*(3), Article 55. https://doi.org/10.1145/3444689